

Experiment-8:

Write a program to implement Linear Regression (LR) algorithm in python

Code:

```
# =====  
# Linear Regression - Medical Insurance Dataset  
# =====  
  
# Upload Dataset  
from google.colab import files  
uploaded = files.upload()  
import pandas as pd  
import io  
  
# Read uploaded CSV  
filename = list(uploaded.keys())[0]  
df = pd.read_csv(io.BytesIO(uploaded[filename]))  
print("First 5 Rows:")  
print(df.head())  
  
# Convert categorical columns into numbers  
from sklearn.preprocessing import LabelEncoder  
le = LabelEncoder()  
df['sex'] = le.fit_transform(df['sex'])  
df['smoker'] = le.fit_transform(df['smoker'])  
df['region'] = le.fit_transform(df['region'])  
  
# Features and Target  
X = df.drop('charges', axis=1)
```

```
y = df['charges']  
# Train-Test Split  
from sklearn.model_selection import train_test_split  
X_train, X_test, y_train, y_test = train_test_split(  
    X,  
    y,  
    test_size=0.2,  
    random_state=42  
)  
# Linear Regression Model  
from sklearn.linear_model import LinearRegression  
model = LinearRegression()  
model.fit(X_train, y_train)  
# Prediction  
y_pred = model.predict(X_test)  
# Evaluation  
from sklearn.metrics import mean_absolute_error  
from sklearn.metrics import mean_squared_error  
from sklearn.metrics import r2_score  
print("\nMean Absolute Error (MAE):")  
print(mean_absolute_error(y_test, y_pred))  
print("\nMean Squared Error (MSE):")  
print(mean_squared_error(y_test, y_pred))  
print("\nR2 Score:")  
print(r2_score(y_test, y_pred))
```

```
print("\nFirst 10 Predictions:")
```

```
result = pd.DataFrame({  
    "Actual": y_test.values,  
    "Predicted": y_pred  
})
```

```
print(result.head(10))
```

Output:

```
Choose Files insurance[1].csv  
... insurance[1].csv(text/csv) - 55628 bytes, last modified: 8/4/2026 - 100% done  
Saving insurance[1].csv to insurance[1].csv  
First 5 Rows:  
   age  sex  bmi  children  smoker  region  charges  
0   19 female  27.900         0    yes southwest  16884.92400  
1   18  male  33.770         1     no southeast   1725.55230  
2   28  male  33.000         3     no southeast   4449.46200  
3   33  male  22.705         0     no northwest  21984.47061  
4   32  male  28.880         0     no northwest   3866.85520  
  
Mean Absolute Error (MAE):  
4186.508898366433  
  
Mean Squared Error (MSE):  
33635210.431178406  
  
R2 Score:  
0.7833463107364539  
  
First 10 Predictions:  
   Actual  Predicted  
0  9095.06825  8924.407244  
1  5272.17580  7116.295018  
2 29330.98315 36909.013521  
3  9301.89355  9507.874691  
4 33750.29180 27013.350008  
5  4536.25900 10790.779562  
6  2117.33885   226.298446  
7 14210.53595 16942.715999  
8  3732.62510  1056.630794  
9 10264.44210 11267.919973
```